Internet Protocol (IP)

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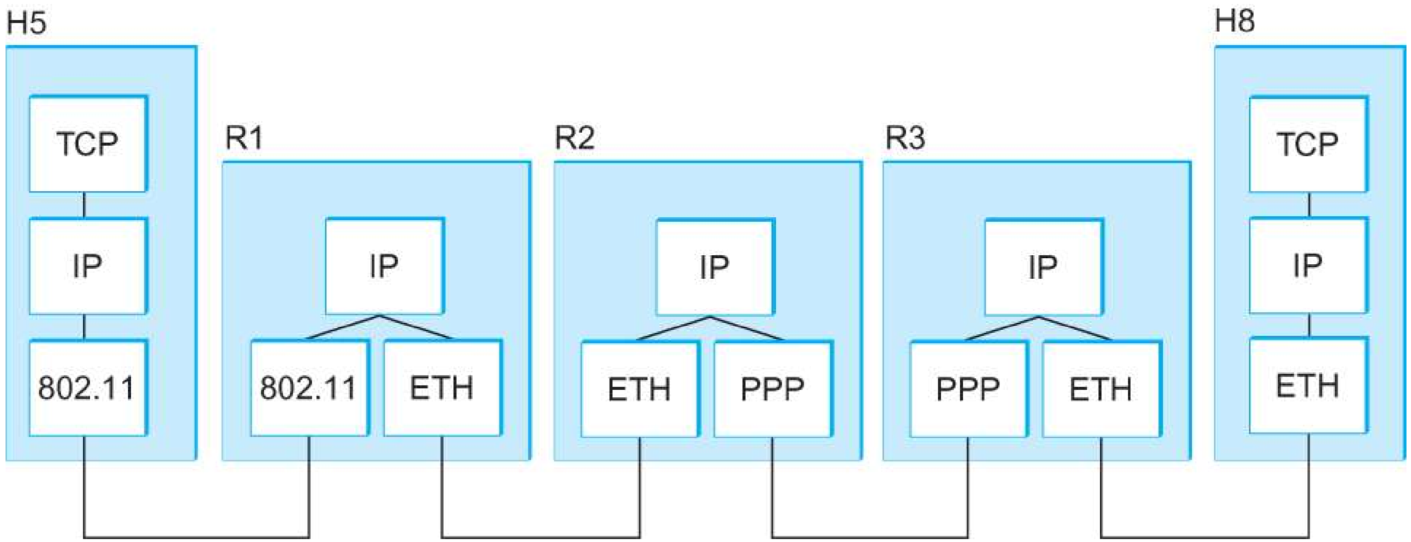
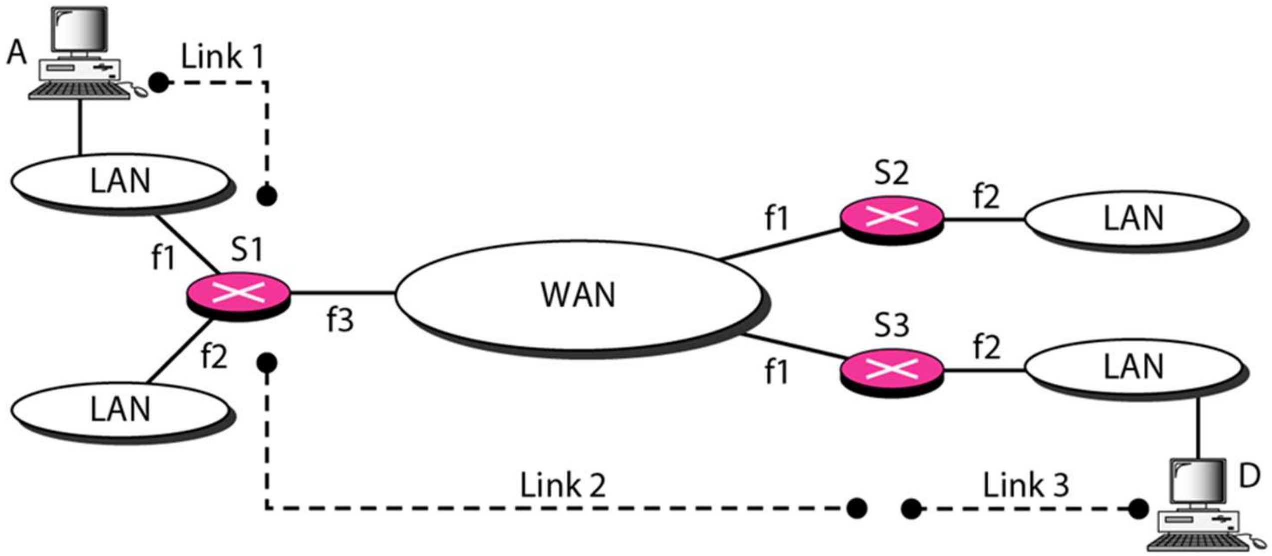
Internet Protocol (IP)

- A Network layer protocol
- Allows delivery of packets across heterogeneous systems



IP: A Network Layer Protocol

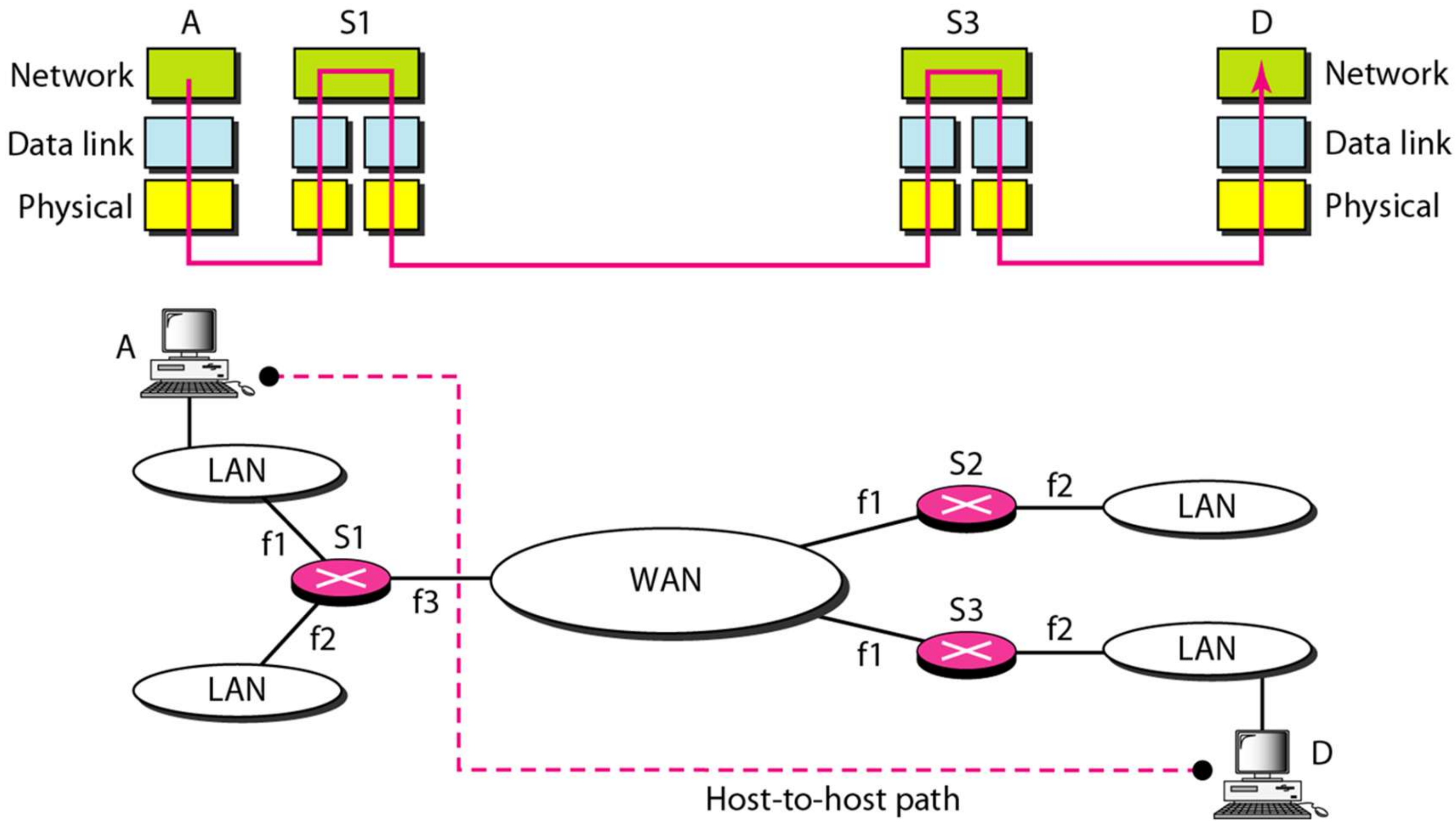
- Packet flow without Network layer



Communication only possible between physically connected hosts

IP: A Network Layer Protocol

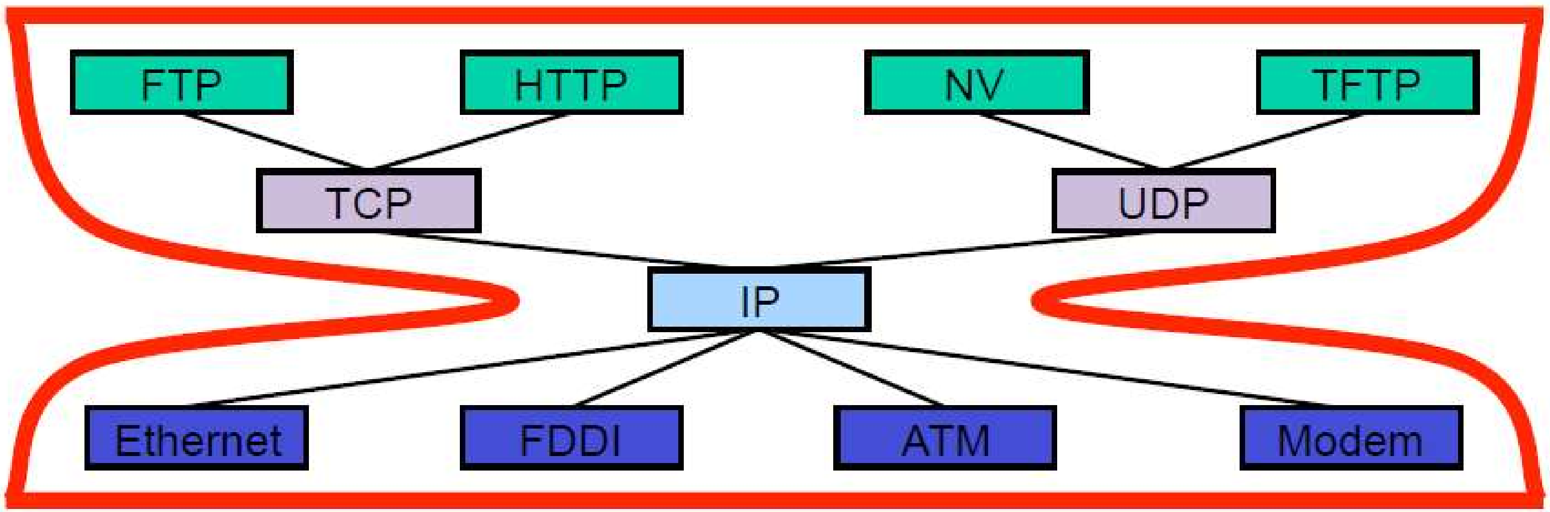
- Packet flow with Network layer



"Multi-hop" communication possible using routers

Hourglass model

- IP is the unified protocol required for all routers to implement
- Networks may implement different transport, data link, and physical layer protocols, but they all must implement IP



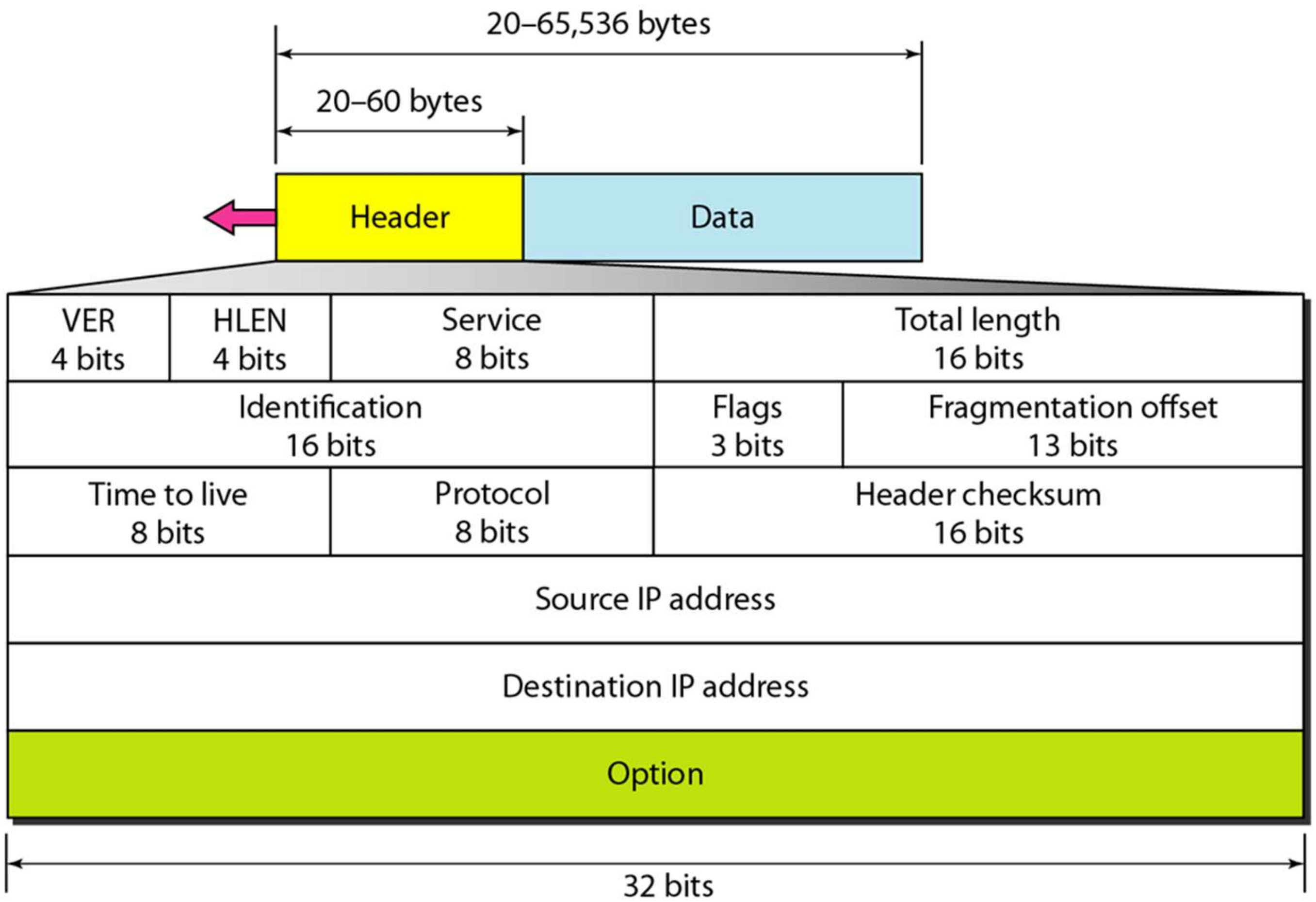
IP: Service Model

- Packet delivery model
  - Connectionless
  - Best-effort (unreliable)
    - Packets may get lost
    - Packets can be delivered out-of-order
    - Duplicate copies may be delivered
    - Packets can be delayed for a long time

- Global Addressing Scheme
  - Provides a way to identify all hosts in the network

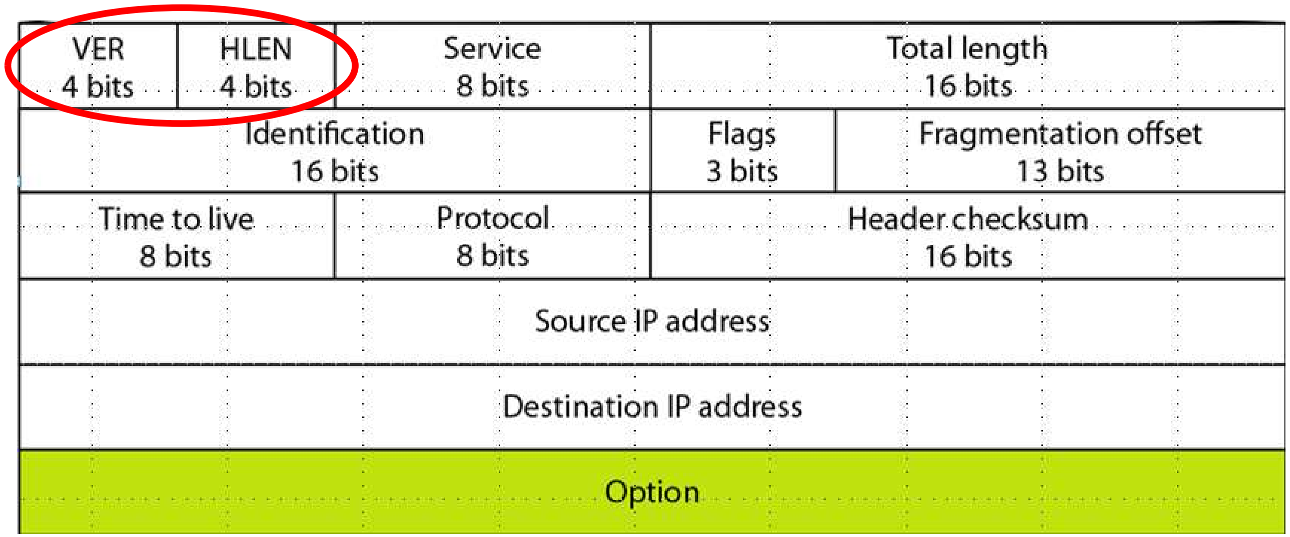
IP: Packet Format

- IP Header is attached in front of the packet
- 20 bytes with no option, up to 60 bytes with option



IP header

- VER: IPv4 or IPv6
- HLEN: length of the header
  - unit: word = 4 bytes
  - If the header is 20 bytes, then HLEN = 5



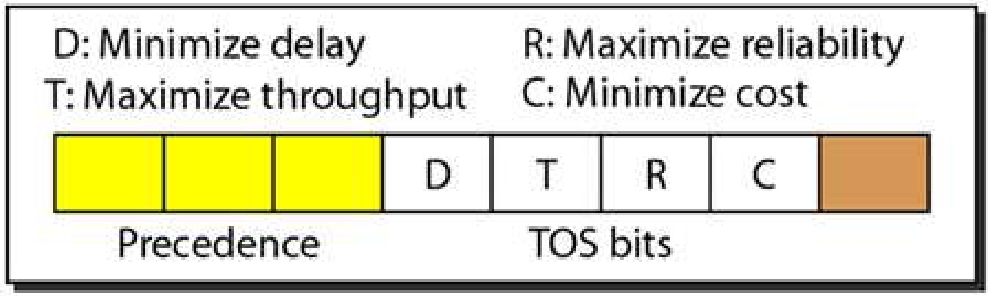
IP header

- Service
  - Service field (8 bits) determines how the packet is processed at routers
  - This field has evolved over time
    - Originally defined as TOS (Type of Service) in RFC 791 (1981)
    - Redefined as DSCP + ECN in RFC 2474 / 3168 (1998-2001)

<table>
<tr><td colspan="4">VER
4 bits</td><td colspan="4">HLEN
4 bits</td><td colspan="8">Service
8 bits</td><td colspan="16">Total length
16 bits</td></tr>
<tr><td colspan="16">Identification
16 bits</td><td colspan="3">Flags
3 bits</td><td colspan="13">Fragmentation offset
13 bits</td></tr>
<tr><td colspan="8">Time to live
8 bits</td><td colspan="8">Protocol
8 bits</td><td colspan="16">Header checksum
16 bits</td></tr>
<tr><td colspan="32">Source IP address</td></tr>
<tr><td colspan="32">Destination IP address</td></tr>
<tr><td colspan="32">Option</td></tr>
</table>

Service class: Original TOS (Historical)

- Original TOS format (RFC 791, 1981)
  - Precedence (3 bits): priority 0-7 (7 = highest)
    - Router drops the lowest-priority packets first when overloaded
  - TOS bits (4 bits): D / T / R / C
    - D: minimize delay
    - T: maximize throughput
    - R: maximize reliability
    - C: minimize cost
    - Only one bit can be 1 at a time (or all 0)

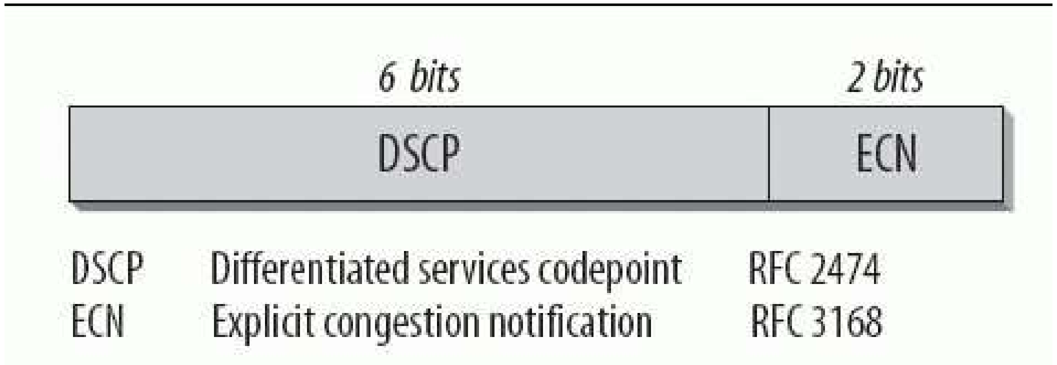


Service type

Service class: DSCP and ECN (Modern)

- DSCP (Differentiated Services Code Point)
  - Defines per-hop behavior (PHB) for Quality of Service
  - Routers can prioritize packets based on DSCP value

- ECN (Explicit Congestion Notification)
  - Allows routers to signal congestion without dropping packets
  - Sender can react before packet loss occurs



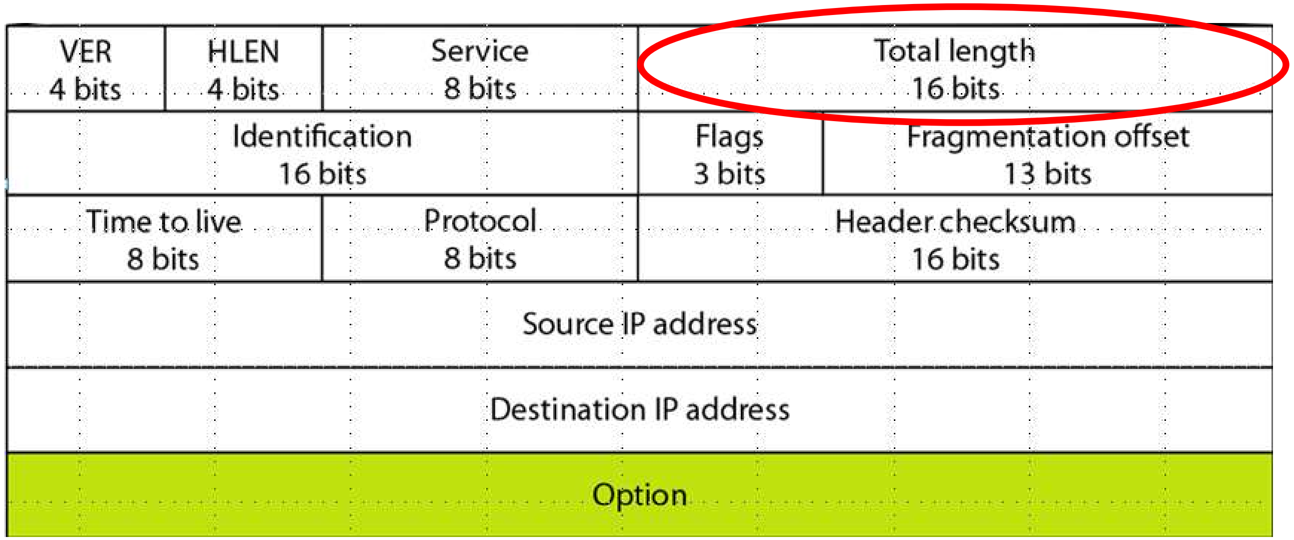
DSCP Values

- DSCP defines standardized per-hop behavior (PHB) classes
  - Each class has a recommended treatment
- EF (Expedited Forwarding)
  - Higest priority: used for VoIP, real-time video
- AF (Assured Forwarding)
  - 4 classes x 3 drop priorities - streaming, video conferencing
- CS (Class Selector)
  - backward-compatible with old precedence values
- Default
  - best-effort (most Internet traffic)

Decimal	DSCP	Description
0	Default	Best Effort
8	CS1	Class 1 (CS1)
10	AF11	Class 1, Gold (AF11)
12	AF12	Class 1, Silver (AF12)
14	AF13	Class 1, Bronze (AF13)
16	CS2	Class 2 (CS2)
18	AF21	Class 2, Gold (AF21)
20	AF22	Class 2, Silver (AF22)
22	AF23	Class 2, Bronze (AF23)
24	CS3	Class 3 (CS3)
26	AF31	Class 3, Gold (AF31)
28	AF32	Class 3, Silver (AF32)
30	AF33	Class 3, Bronze (AF33)
32	CS4	Class 4 (CS4)
34	AF41	Class 4, Gold (AF41)
36	AF42	Class 4, Silver (AF42)
38	AF43	Class 4, Bronze (AF43)
40	CS5	Class 5 (CS5)
46	EF	Expedited Forwarding (EF)
48	CS6	Control (CS6)
56	CS7	Control (CS7)

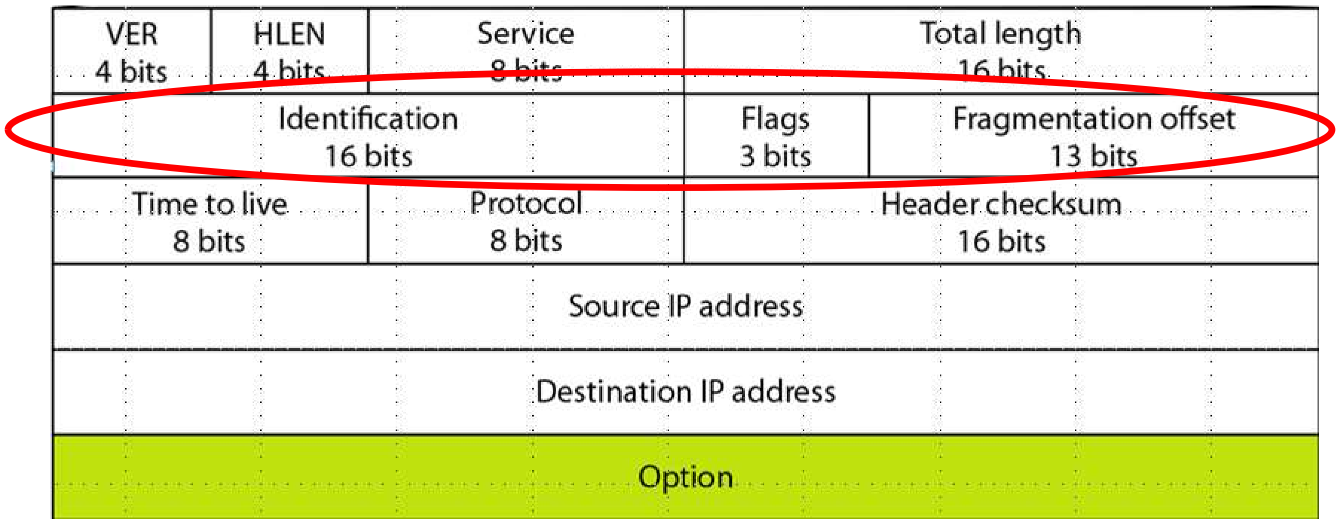
IP header

- Total length: the total packet length (header + data)
  - <u>unit</u>: byte



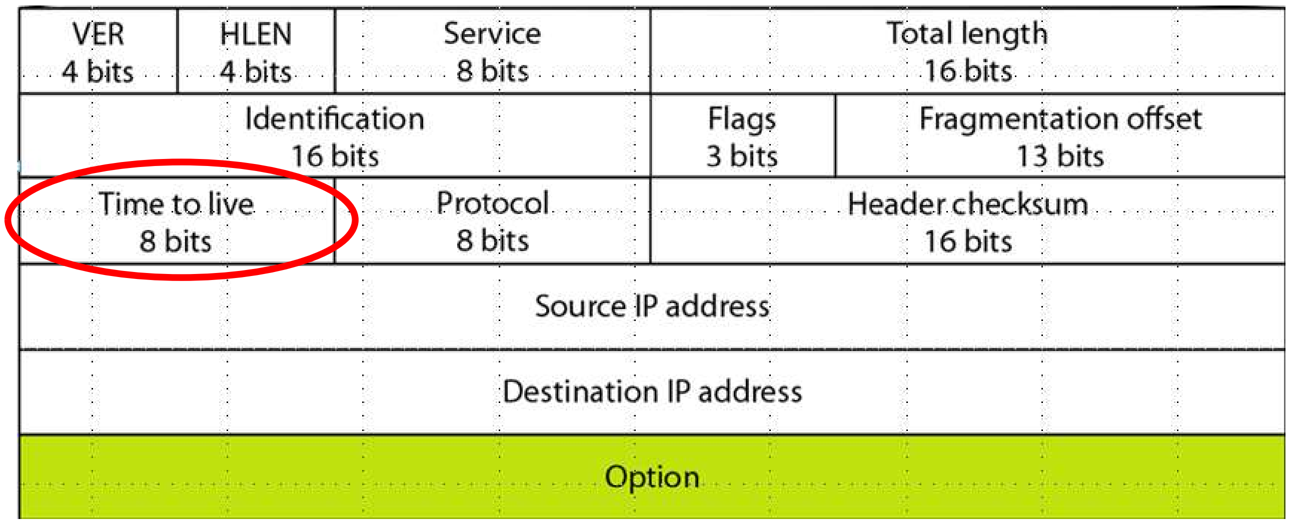
IP header

- Identification, Flags, Fragmentation offset
- Related to fragmentation and assembly
  - Explained later



IP header

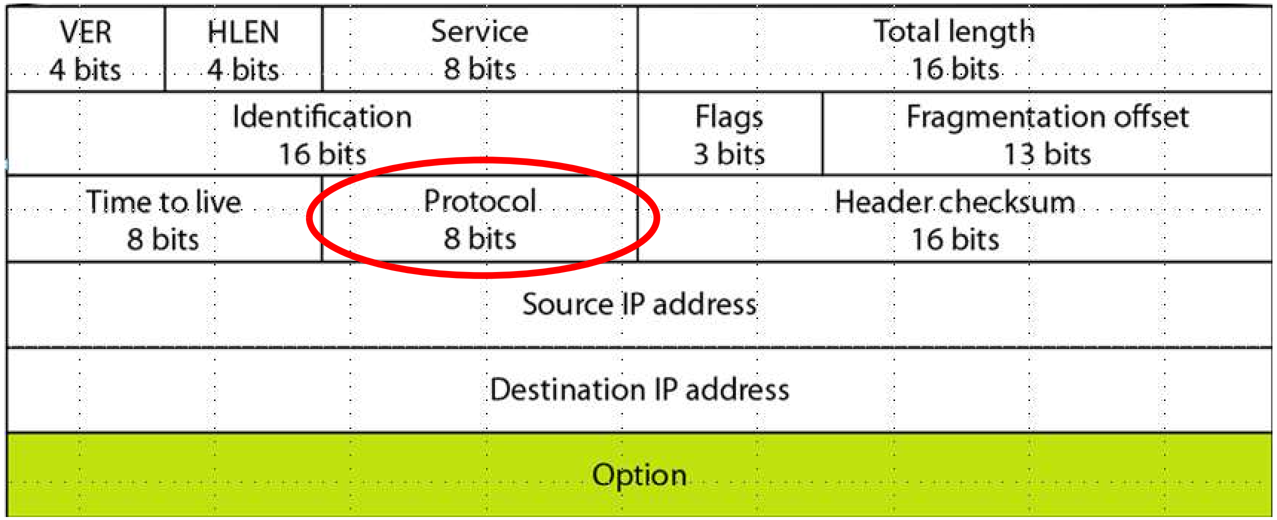
- Time to live: the life of the packet
  - unit: number of hops (8 bits → max 255)
  - decremented by 1 at each router
  - when TTL becomes 0, the router drops the packet
    - Prevents packets from looping forever in case of routing errors
    - When the packet is dropped, an ICMP packet is generated at the router and sent to the packet source



IP header

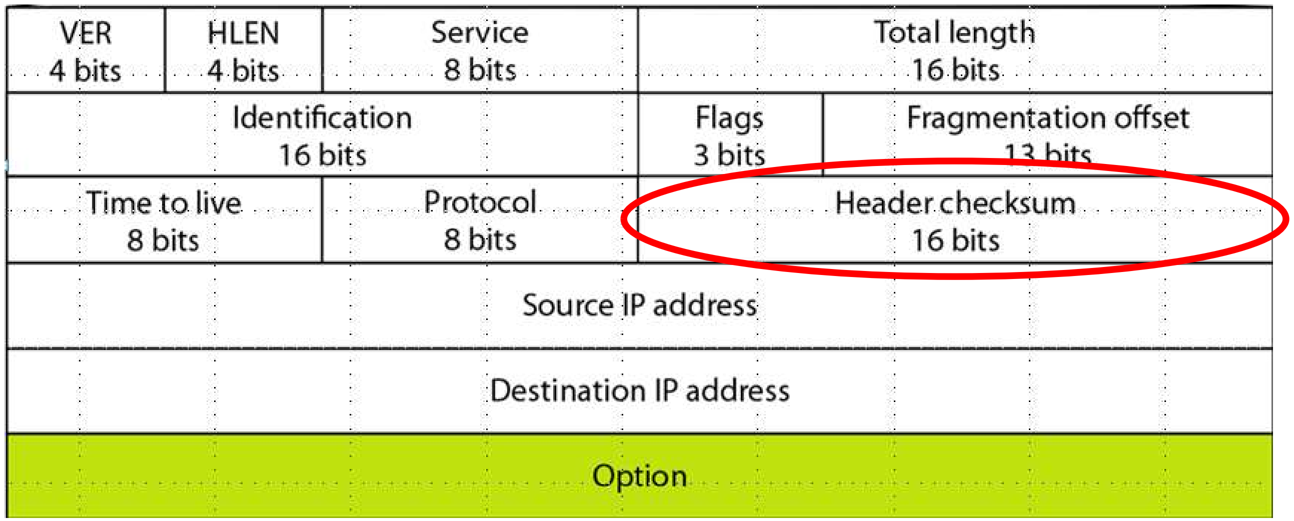
- Protocol: protocol of the upper layer (transport)
  - used for demultiplexing to the correct handler at the received

Value	*Protocol*
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF



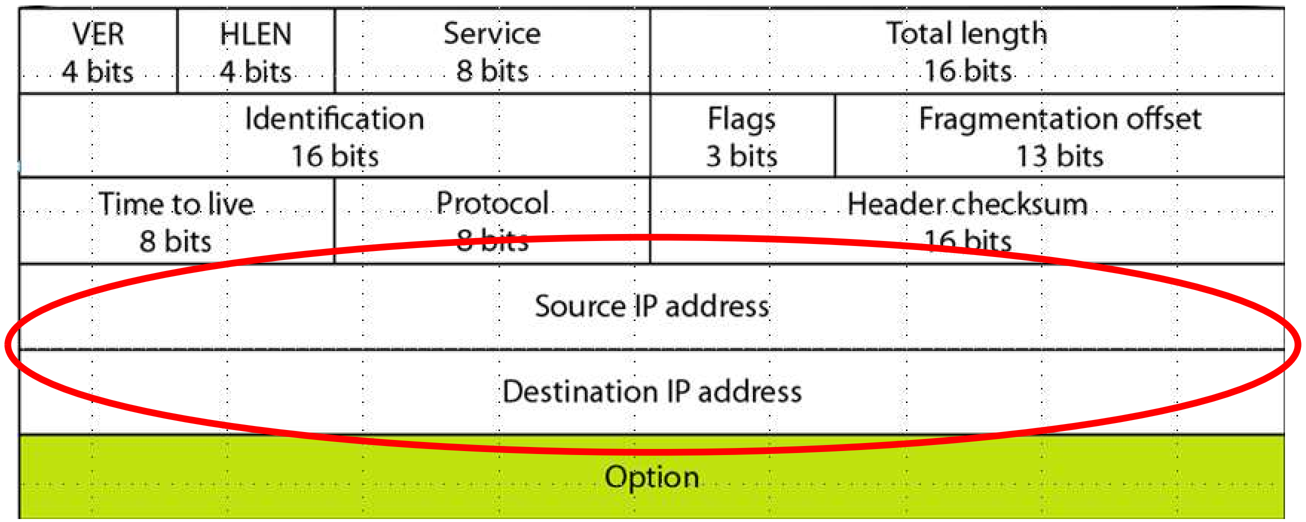
IP header

- Header checksum: used for error detection



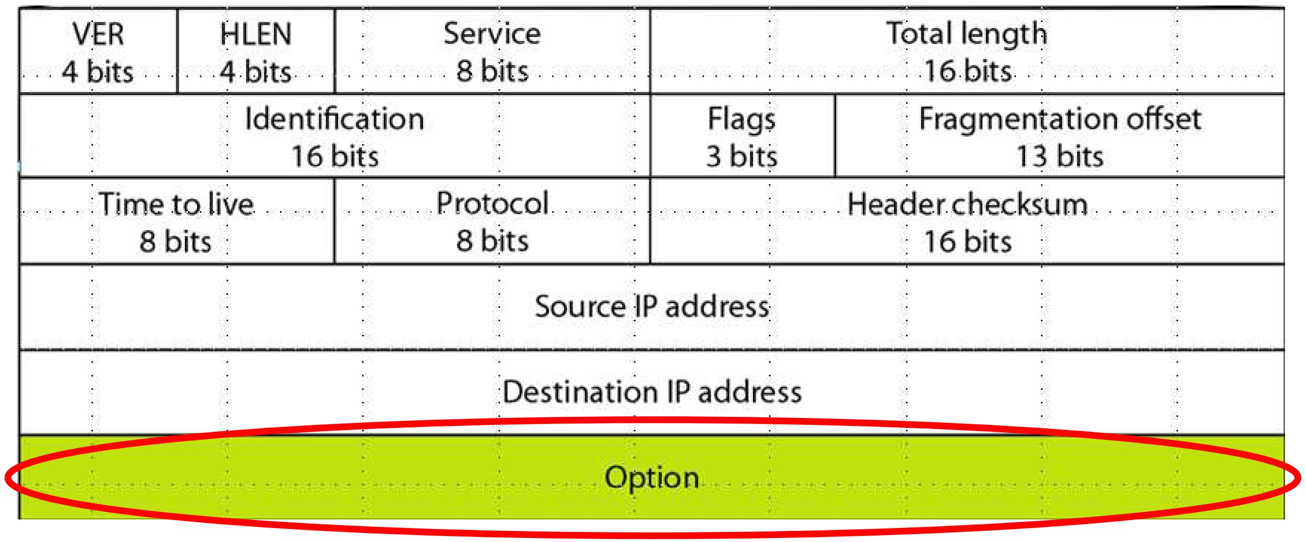
IP header

- Source IP address (32 bits)
- Destination IP address (32 bits)



IP header

- Option: optional, may not be present
  - Other 20 bytes are mandatory fields
  - Thus, the minimum HLEN is 5
  - When option fields are present, HLEN is larger than 5
  - Examples: record route, timestamp, source routing
  - rarely used in practice



IP header: exercise

- The initial 8 bits of an IPv4 packet was the following

  01000010

- The router drops the packet since it is an erroneous packet. Why?

IP header: exercise

- HLEN value of an IPv4 packet is 1000 (in binary). What is the length of option fields in its IP header?

IP header: exercise

- HLEN of a packet is 5, and the total length field has the value 0x0028. What is the length of <u>data</u> in this IP packet?

IP header: exercise

- The initial part of an IPv4 packet is like the following:

0x45000028000100000102

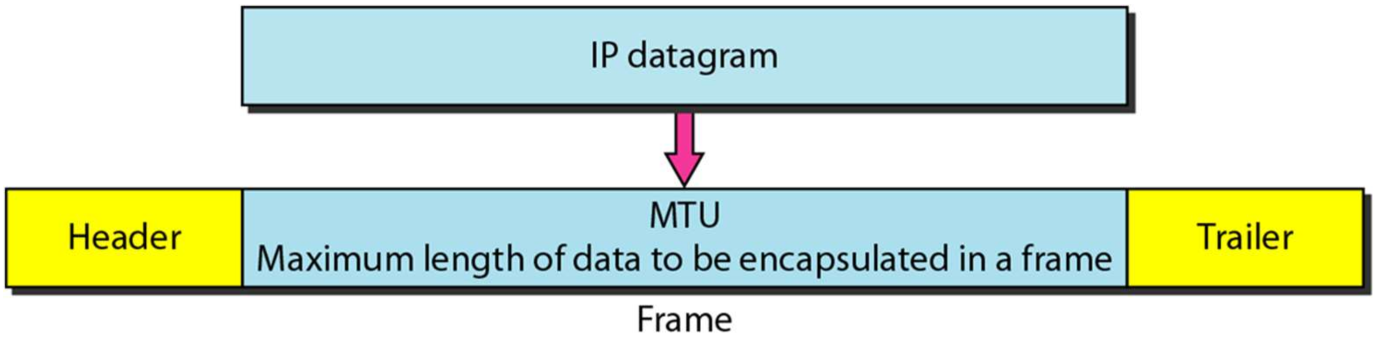
- What is the time-to-live of this packet?

MTU (Maximum Transfer Unit)

- Every data link layer protocol has an upper limit of IP packet

network layer

data link layer



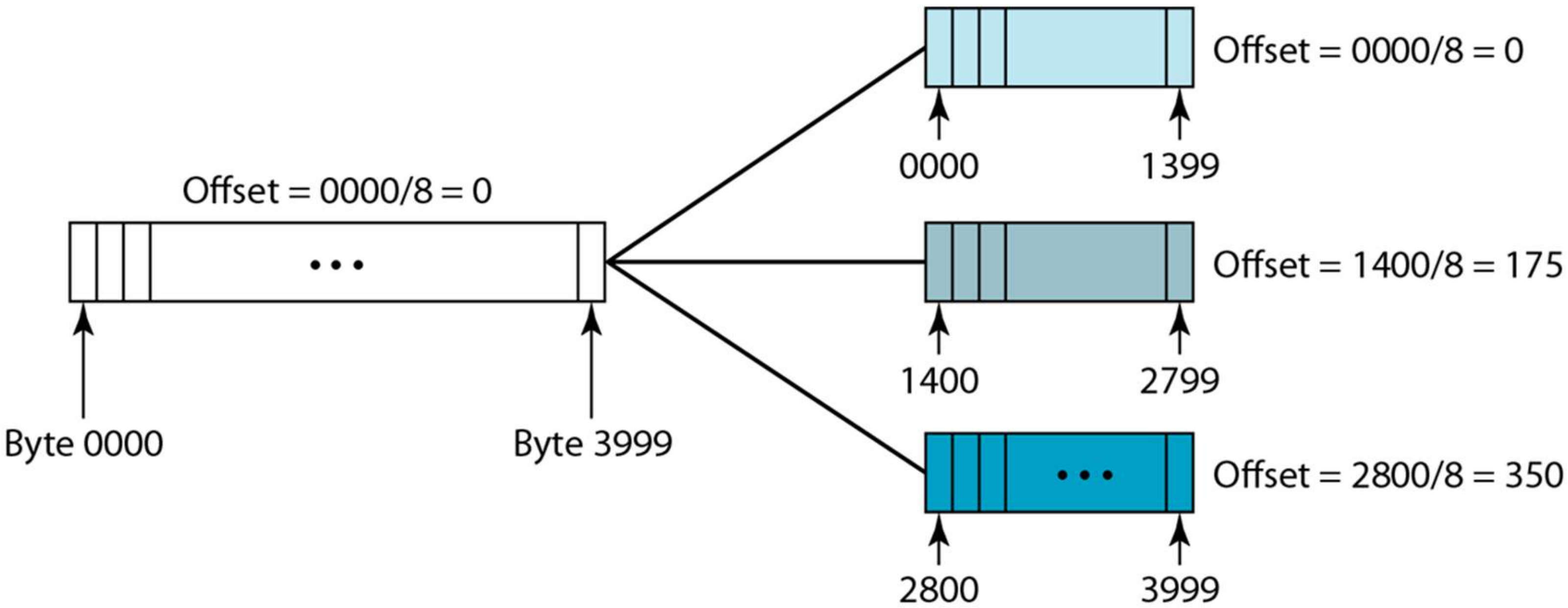
MTU (Maximum Transfer Unit)

- Different protocols have different MTUs
  - MTU depends on forwarding and error detection methods

Protocol	*MTU*
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

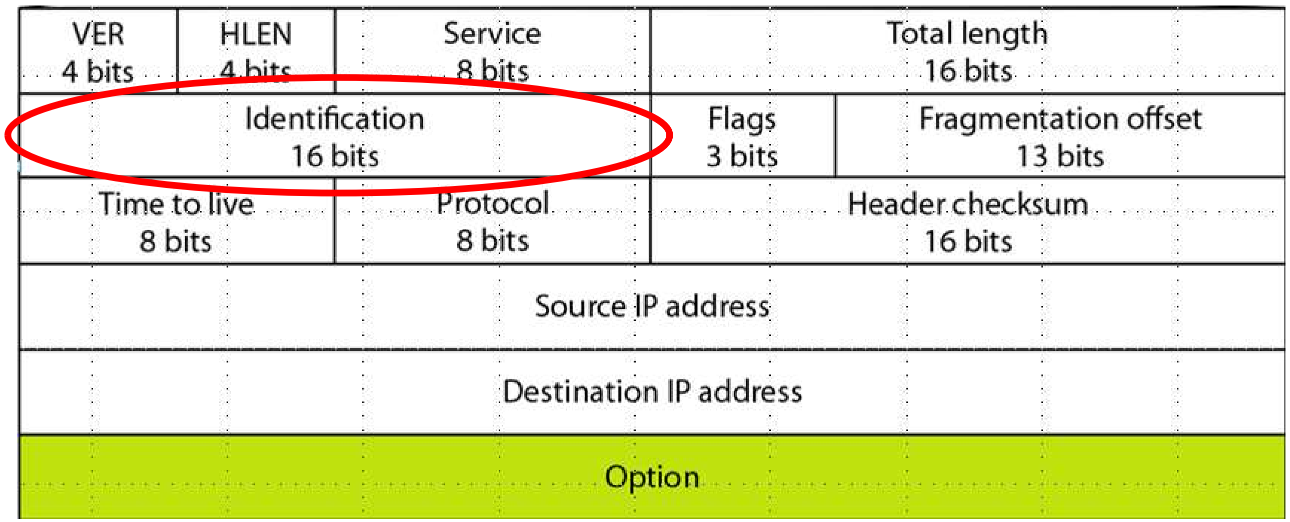
Fragmentation

- If size of an IP packet is larger than MTU, then the packet needs to be fragmented into multiple packets
  - The IP header is attached to each fragment



Fields related to Fragmentation

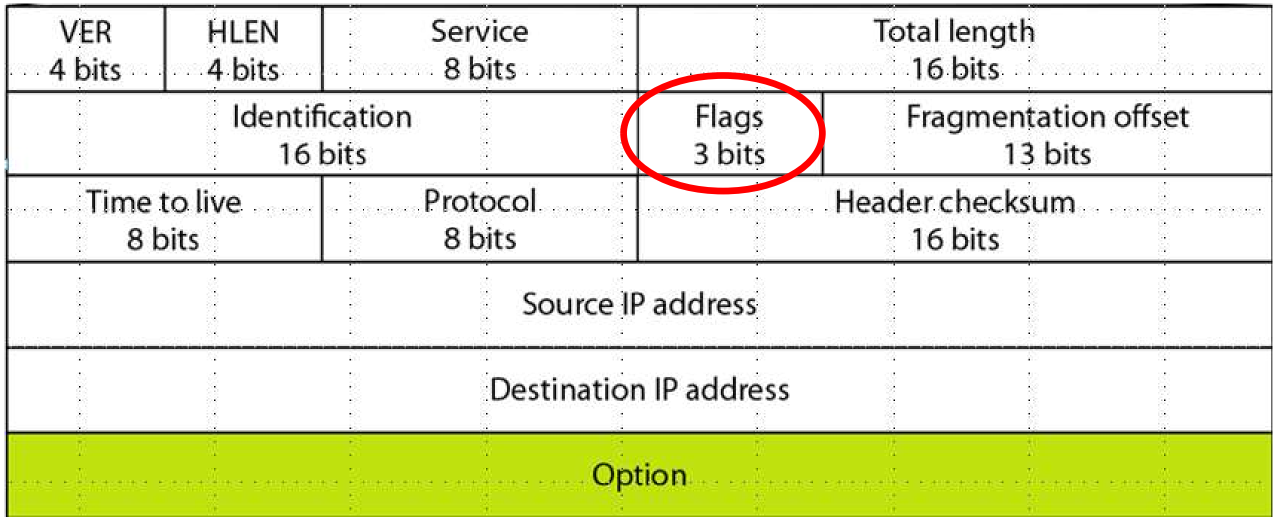
- Identification: packet identifier
  - All fragments of the same packet have the same identification number



Fields related to Fragmentation

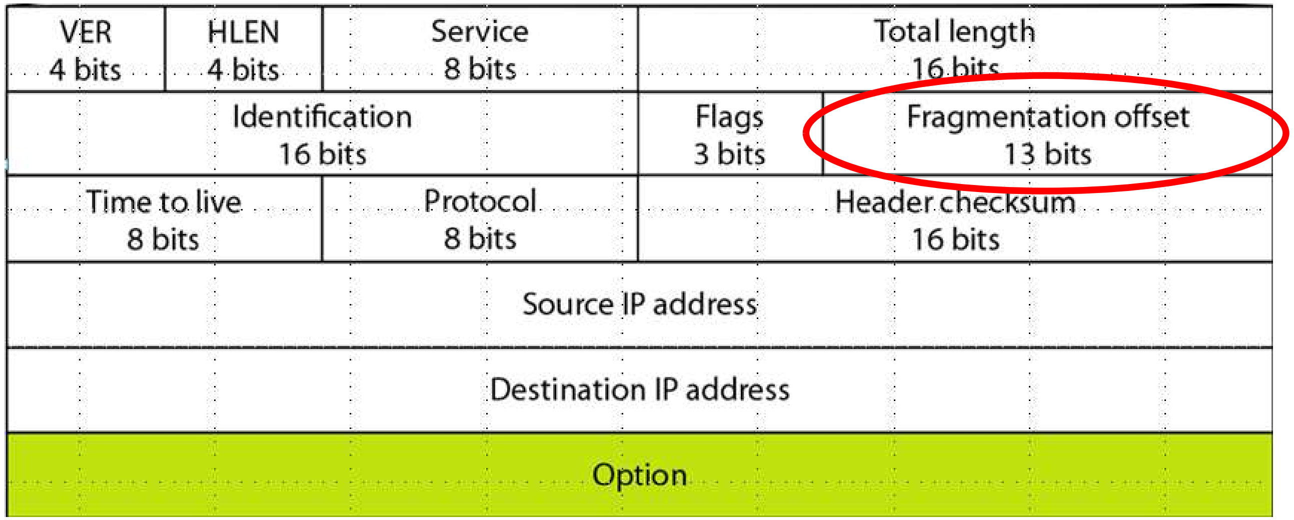
- Flag
  - If the bit 'D' is 1: this packet should not be fragmented
    - If packet size is larger than MTU, packet is discarded
  - If the bit 'M' is 1: there are more fragments after this
    - If 'M' is 0, this is the <u>last fragment</u>, or this is the <u>only fragment</u>
  - The first bit is reserved and not used (always set to 0)

D: Do not fragment
M: More fragments

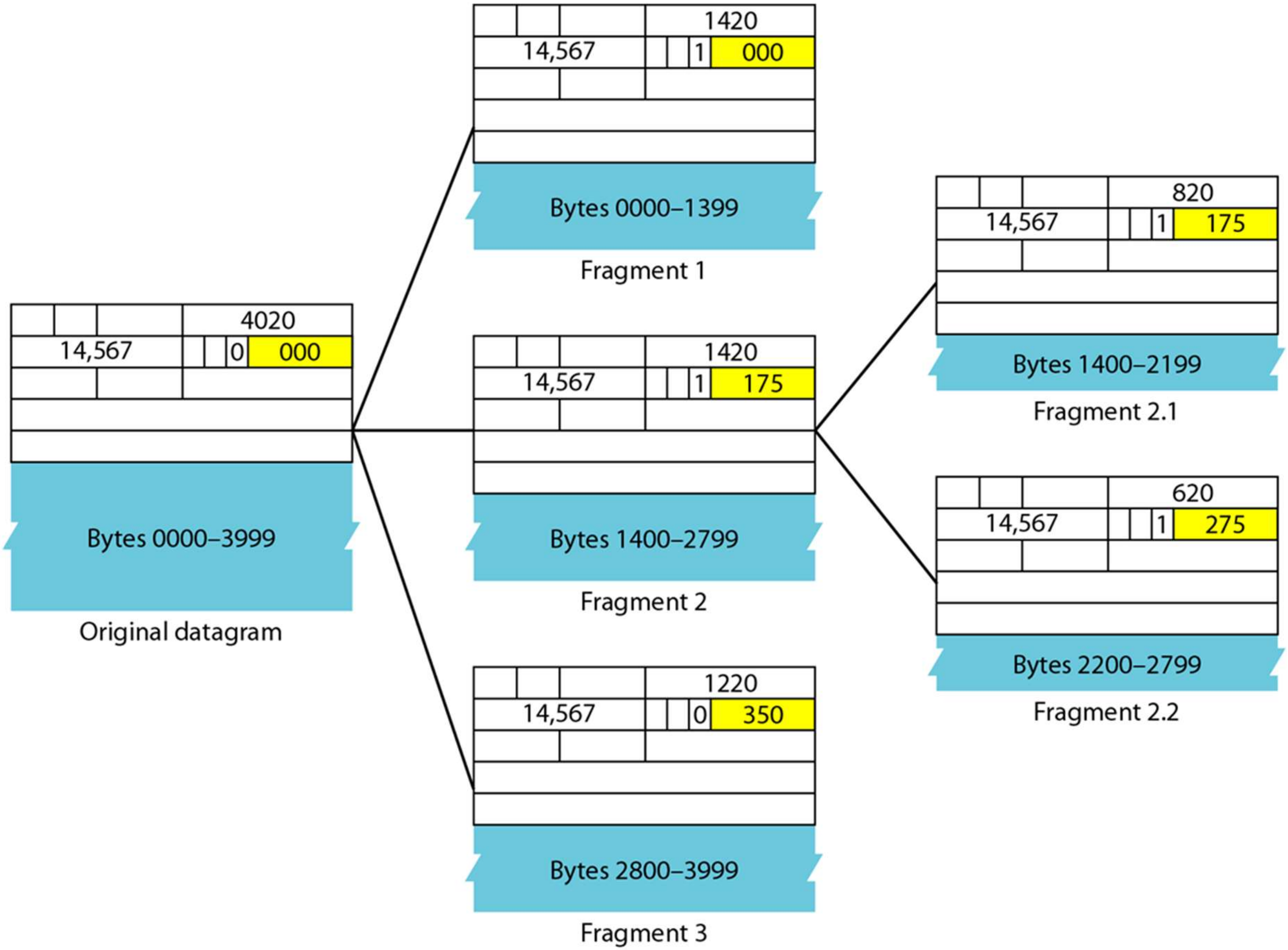


Fields related to Fragmentation

- Fragmentation offset
  - The position of this fragment in the original packet
    - unit: 8 bytes
    - only considers data bytes (refer to the figures in the next slides)

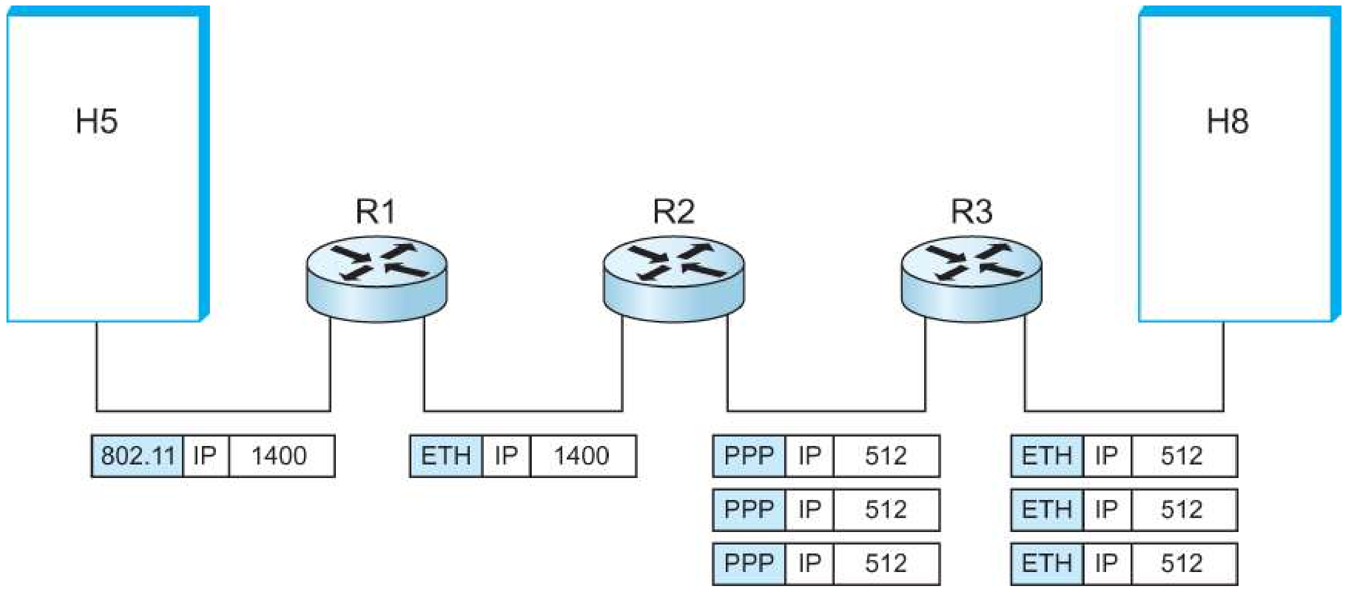


IP fragmentation: example



Fragmentation & reassembly

- Fragmentation can take place at any node along the path
  - Each router fragments independently based on the next-link MTU
- Reassembly only occurs at the final destination
  - Fragments may take different paths (IP is connectionless)
  - Reassembly here would have to be undone at the next small link
- The whole packet is discarded if even one fragment is lost
  - Losing 1 of N fragments wastes all the routers



Fragmentation & reassembly: exercise

- A packet has arrived, and its 'M' bit is 0. What is the meaning of this?

- Answer: either of the following
  - This packet is not fragmented
  - This packet is fragmented, and this fragment is the last one

Fragmentation & reassembly: exercise

- A packet has arrived, and its 'M' bit is 1. What is the meaning of this?

- Answer
  - This packet is fragmented, and this fragment is not the last one

Fragmentation & reassembly: exercise

- A packet has arrived, and its 'M' bit is 1, and the fragmentation offset is 0. What is the meaning of this?

- Answer
  - This packet is fragmented and this is the first fragment (there are more fragments after me)

Fragmentation & reassembly: exercise

- A packet has arrived, and its fragmentation offset is 100. What is the meaning of this?

- Answer
  - This packet is fragmented, and the position of this fragment is 800 bytes (only counting data bytes)

Fragmentation & reassembly: exercise

- A packet has arrived. Its fragmentation offset is 100, HLEN is 5, and total length is 100. What is the first and last position of this fragment in the original packet? (the first and the last bytes)

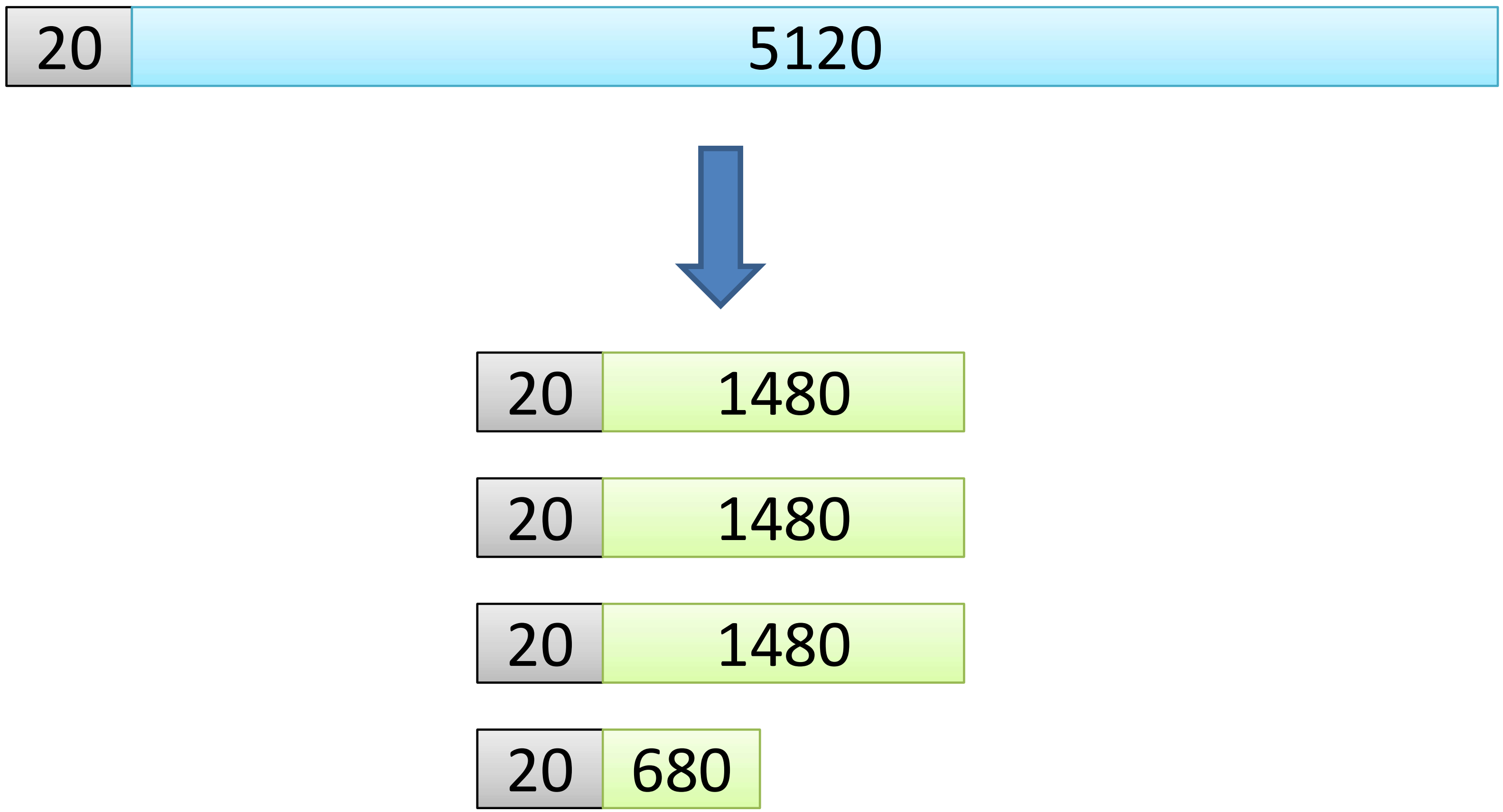
- the first byte: 100 x 8 = byte #800
- the last byte
  - total length: 100 bytes
  - header length: 20 bytes
  - data length: 100 – 20 = 80 bytes
  - Thus, the last byte is the byte #879

Fragmentation & reassembly: exercise

- Assume IP header is always 20 bytes
- An IP packet has a size of 5140 bytes (including header)
- If data link layer MTU is 1500 bytes, then how should the packet be fragmented?

Fragmentation & reassembly: exercise

- The packet should be fragmented as follows.
  - MTU = 1500



Path MTU Discovery

- Modern hosts find the smallest MTU on the path beforehand, so no router needs to fragment
- Mechanism
  - Sender sets DF flag on every packet
  - If a router cannot forward, it drops the packet and replies with ICMP "Fragmentation Needed" (with the next-hop MTU)
  - Sender reduces packet size and retries
- Benefits
  - No "one lost fragment kills the whole packet" problem